

Model Generalization and Data Preparation in Machine Learning

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Machine Learning

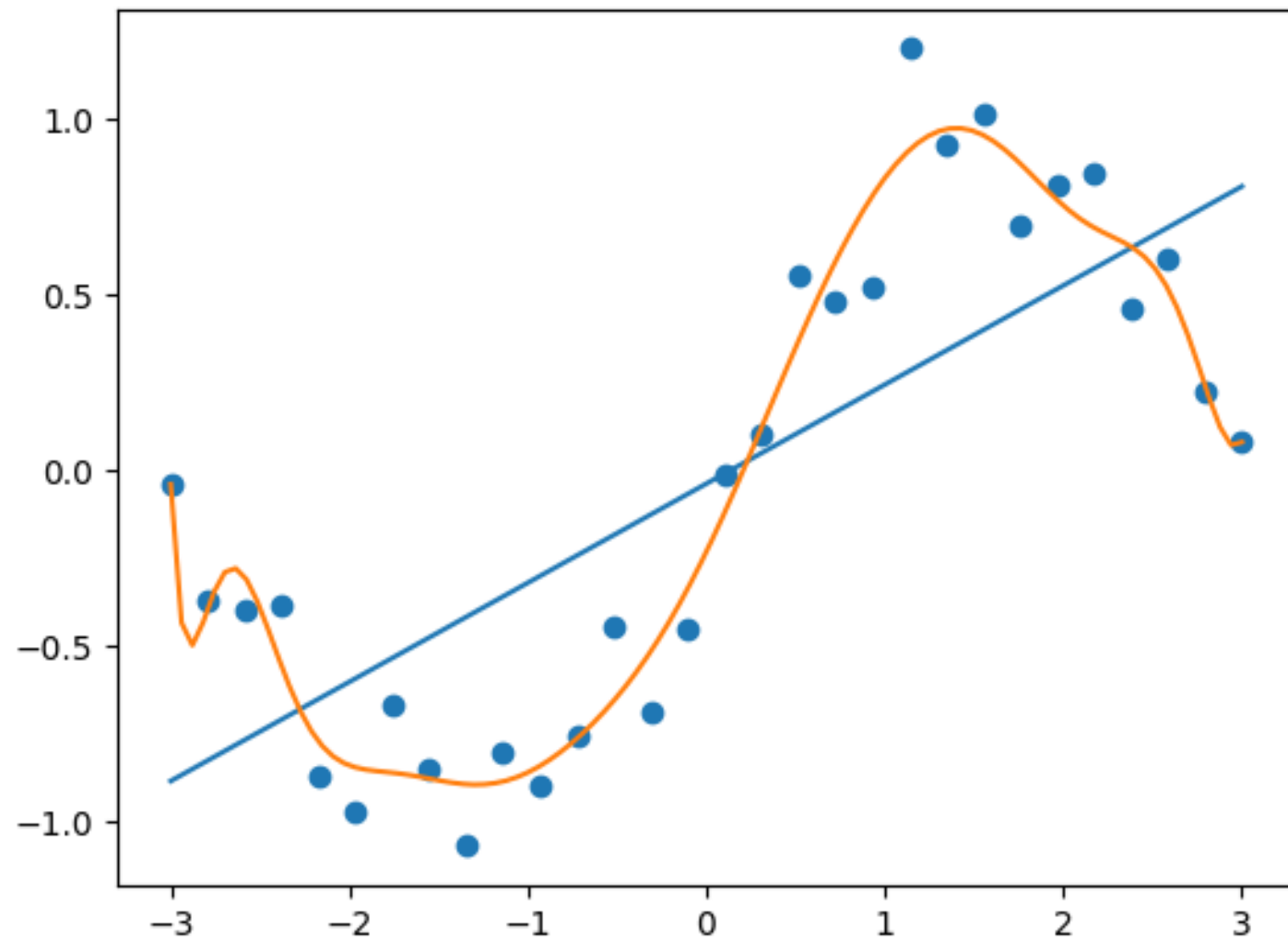
Content

- Overfitting & Underfitting
- Bias-Variance Tradeoff
- Data Preprocessing
- Feature Selection

Overfitting: Model captures noise and performs poorly on test data.

- Model learns training data too well
- Captures noise and random fluctuations
- High training accuracy
- Low test accuracy

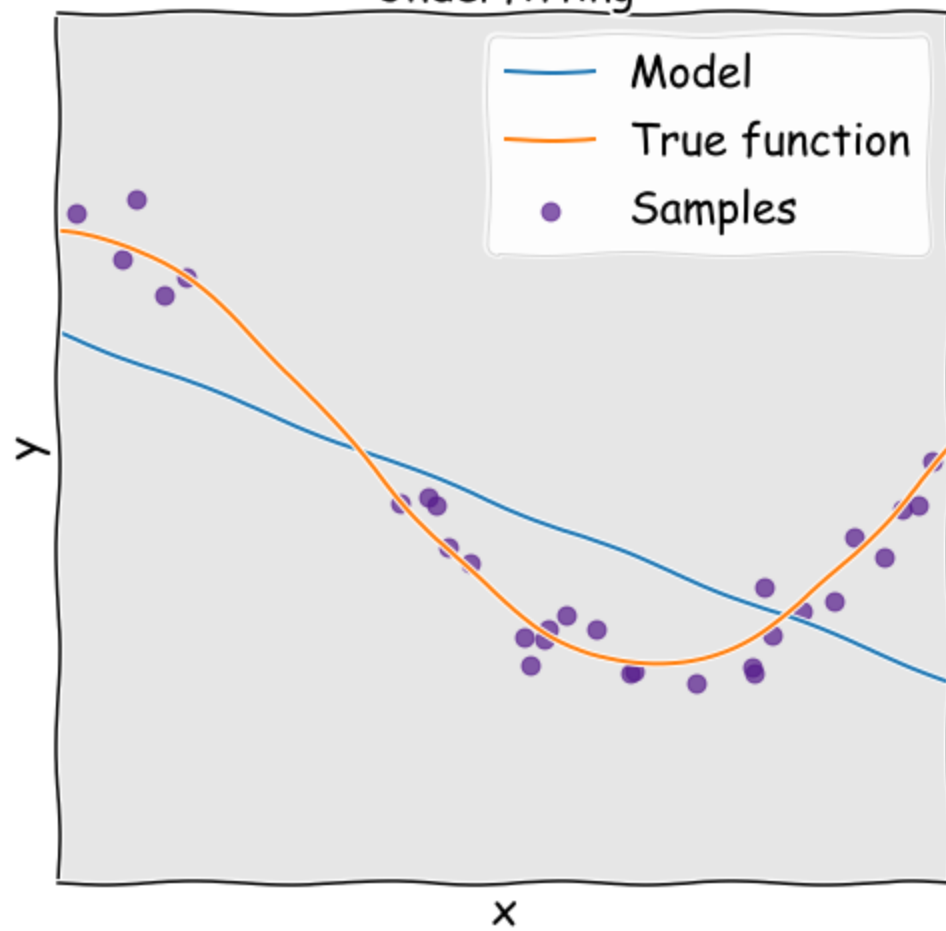
Overfitting vs Underfitting



Underfitting: Model too simple to capture pattern.

- Model too simple
- Cannot capture underlying pattern
- Low training accuracy
- Low test accuracy

Underfitting



Comparison of Overfitting and Underfitting.

Aspect	Overfitting	Underfitting
Model Complexity	High	Low
Training Error	Low	High
Testing Error	High	High
Generalization	Poor	Poor

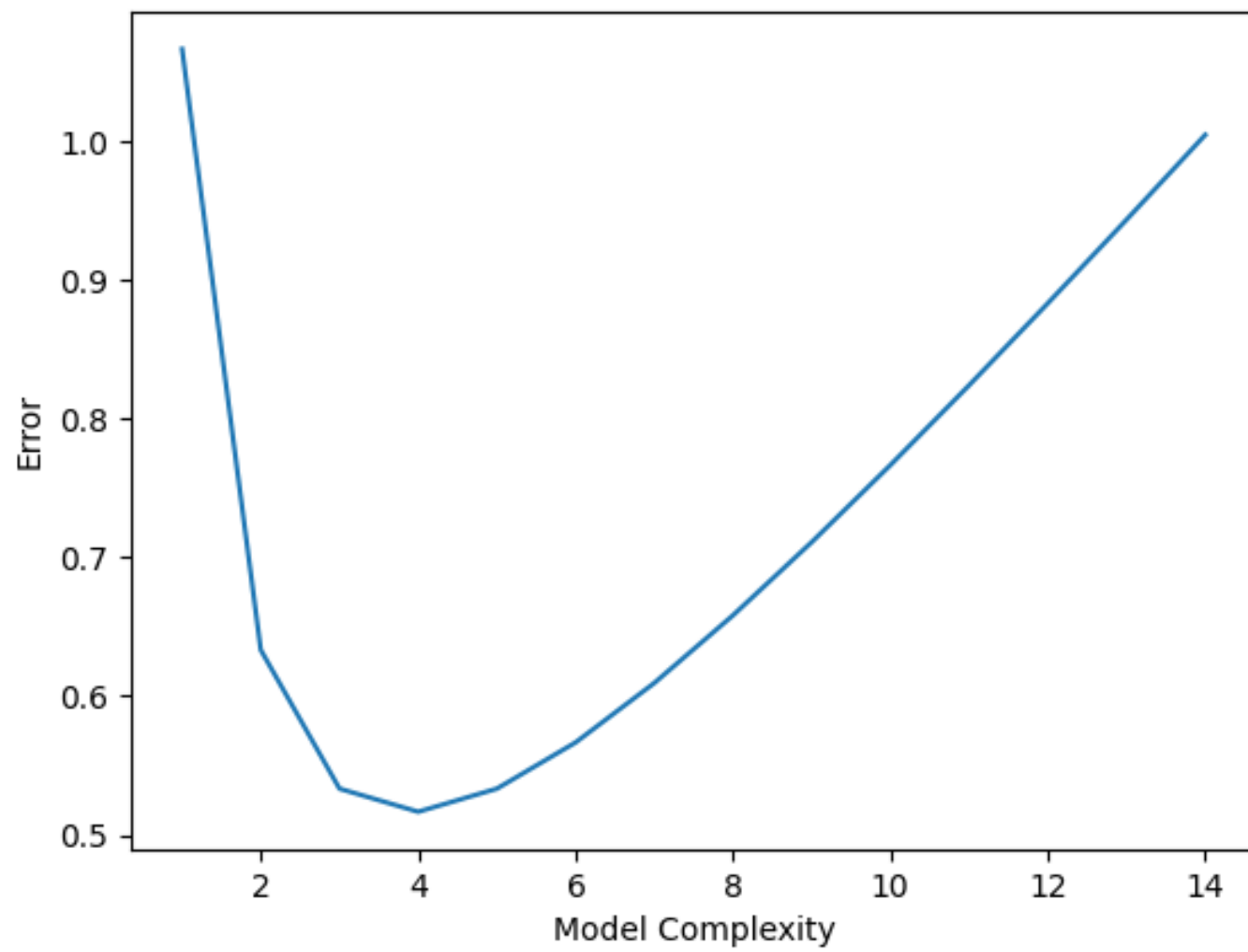
Bias-Variance Tradeoff Concept.

Bias $\rightarrow$ Error from wrong assumptions

Variance $\rightarrow$ Error from sensitivity to fluctuations

Tradeoff between the two

Bias-Variance Tradeoff



Total Error = Bias² + Variance + Irreducible Error.

- Total Error = Bias² + Variance + Irreducible Error
 - Bias² → Systematic error
 - Variance → Model instability
 - Irreducible error → Noise in data

Techniques to prevent overfitting.

- Cross-validation
- Regularization (L1, L2)
- Pruning (for decision trees)
- Dropout (for neural networks)
- Early stopping

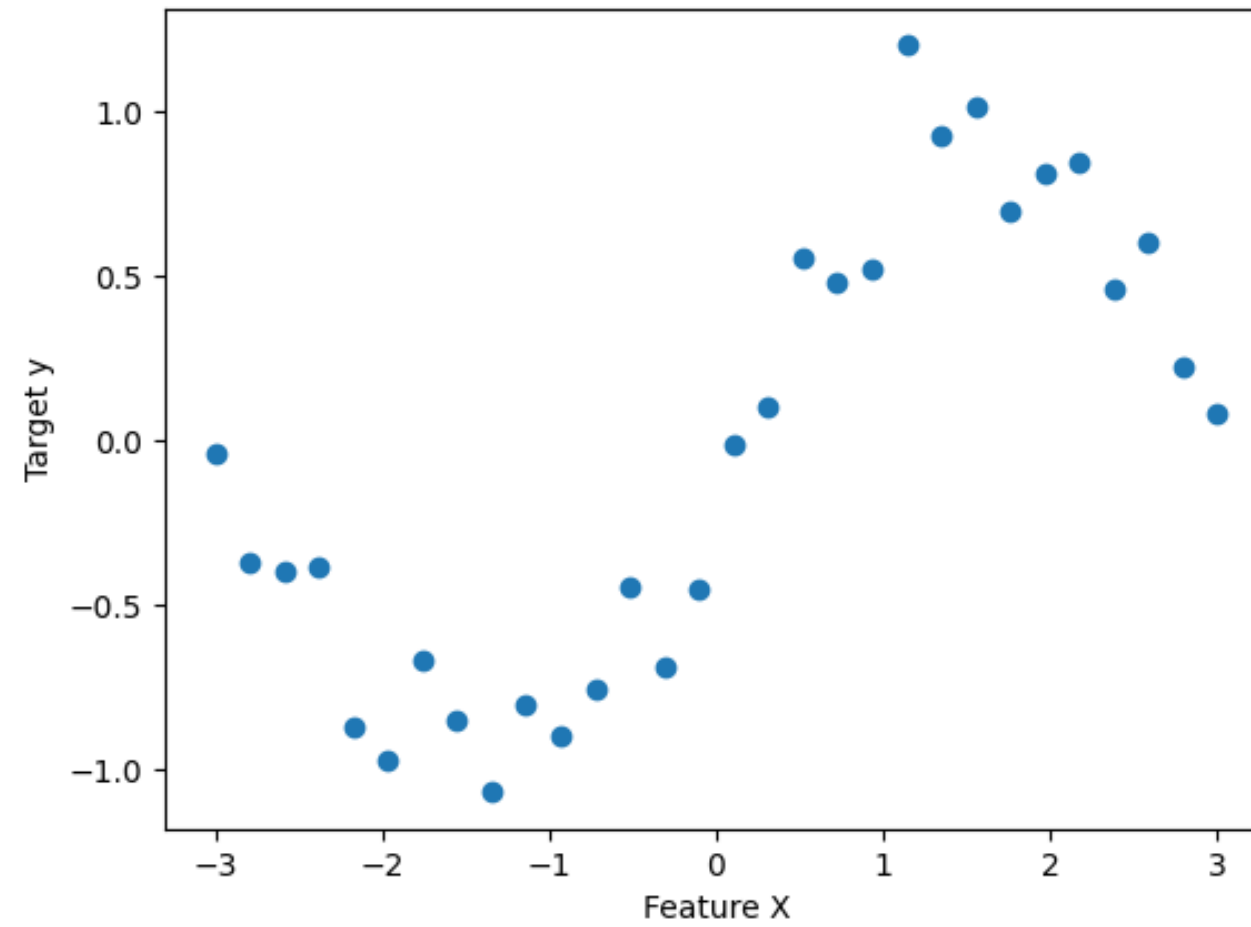
Importance of Data Preprocessing.

- Why preprocessing is important:
- Real-world data is messy
- Missing values
- Different scales
- Irrelevant features

Handling Missing Data techniques.

- Common Techniques:
- Remove rows
- Mean/Median imputation
- Mode imputation
- Predictive imputation
 - **Example:**
Replace missing salary values with median salary.

Example Dataset



Normalization and Standardization methods.

- Why scaling is required:
 - Algorithms like KNN, SVM are distance-based
 - Features with large ranges dominate
- Methods:
 - Min-Max Scaling
 - Standardization (Z-score)

– **Formula (Z-score):**

$$z = \frac{x - \mu}{\sigma}$$

Feature Selection techniques.

- Why Feature Selection?
 - Reduce dimensionality
 - Improve performance
 - Reduce overfitting
 - Faster training
- Methods:
 1. Filter Methods (Correlation, Chi-square)
 2. Wrapper Methods (Recursive Feature Elimination)
 3. Embedded Methods (LASSO, Decision Trees)

Practical Activity and Discussion.

- **Activity 1 – Dataset Cleaning**
 - Identify missing values
 - Apply imputation
 - Normalize features

Activity 2 – Overfitting Visualization

- Train polynomial regression with degree 1, 5, 15
- Compare training vs testing error
- Plot model curves
- **Discussion Questions:**
 - Which model generalizes best?
 - How does complexity affect variance?